

# JOSE ANTONIO ELIZALDE

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## Education

University of California, Los Angeles

Sept 2025 – June 2027

*B.S. in Computer Science and Engineering*

*Westwood, CA*

## Relevant Coursework

Data Structures and Algorithms, Networking Fundamentals, Distributed Systems, Operating Systems Architecture, Systems Programming, Computer Architecture, Discrete Mathematics, Object-Oriented C++

## Technical Skills

**Languages:** C++, C, C#, Python, Go, TypeScript, SQL, Bash

**Backend & Infrastructure:** Linux, FastAPI, PostgreSQL, Prisma, Docker Compose, Nginx, systemd, DigitalOcean

**Web & Data:** Next.js, React, Tailwind CSS, pandas, yfinance

**Tools:** Git/GitHub, CMake, Unity, Vim/NeoVim, Arduino, LaTeX

## Research

Carnegie Mellon University – REUSE Research Program

May 2025 – Aug 2025

*Undergraduate Researcher, Human-Computer Interaction Institute (HCII)*

*Pittsburgh, PA*

- Developed an interactive application to investigate methods for improving communication and empathy between rideshare drivers and passengers.
- Collaborated with a faculty advisor, graduate students, and a research partner to design and improve the system.
- Co-authored a paper with the CMU faculty which accepted to CHI 2026 (ACM Conference)

University of California, Los Angeles – SURE C<sup>2</sup> Research Program

June 2024 – August 2024

*Undergraduate Researcher, Department of Mathematics*

*Los Angeles, CA*

- Conducted research in combinatorics on partially ordered sets (posets), studying transfer systems and their patterns.
- Discovered a recursive pattern on the set  $X_{n+}^+$ , and co-authored a research paper that was presented at the UT Arlington.
- Developed a C++ program to automate the counting of all the transfer systems on the poset  $X_{n+}^+$

## Experience/Projects

RobinHood Discord Portfolio Bot | C++, D++, Python, FastAPI, Pandas, NumPy

June 2026 – Present

- Engineered a C++ Discord bot (D++) and Python FastAPI backend to deliver real-time Robinhood portfolio analytics and automated stock screening.
- Architected a decoupled, two-process system via HTTP boundaries to maintain persistent brokerage sessions, preventing re-authentication drops during bot restarts.
- Deployed on a DigitalOcean Linux VPS with systemd for 24/7 uptime, integrating a background Pandas pipeline to analyze S&P 500 momentum and valuation metrics.

DailyCodeforce | Next.js, TypeScript, PostgreSQL, Prisma, Go, Docker, Nginx

2026 – Present

- Developed and self-hosted a full-stack competitive programming platform (Next.js, TypeScript) that automatically curates tiered Codeforces problems with C++ reference solutions.
- Implemented intelligent scheduling logic featuring a 90-day deduplication window and 30-day automated backfill, querying a PostgreSQL database managed via Prisma.
- Containerized the web application, a custom Go Discord bot, and an Nginx reverse proxy using Docker Compose to streamline one-command droplet deployments.

Personal Developer Portfolio (joseelizalde.dev) | React, Next.js, Tailwind CSS

2026

- Engineered a responsive, high-performance developer portfolio to showcase software engineering projects, technical skills, and academic research.
- Architected the frontend UI/UX using modern web technologies, optimizing for cross-device accessibility, Lighthouse performance metrics, and SEO.
- Configured a continuous deployment (CI/CD) workflow connected to GitHub, automating the build process to instantly update my website.

Jump the Gun – UCLA ACM Studio | C#, Unity, HLSL, Git

Oct 2024 – Present

- Led the design and development of a first-person action shooter as Game Director, defining gameplay mechanics and coordinating implementation across the team.
- Implemented a custom physics-based movement controller in C#, replacing Unity's default character controller with manually programmed movement and collision logic.
- Utilized Git branching and pull requests to integrate gameplay features developed concurrently by multiple contributors.

Shellscape | C#, Unity, WebGL

2026

- Shipped a browser-playable action game with a team of five, designed and completed inside a game jam window and released publicly on itch.io.
- Implemented the movement and interaction systems, including the body-possession mechanic the game is built around, and built the tutorial scene with its scripted narration and dialogue boxes.